

Software Requirements Specification

MakeMyTrip Clone

A Frontend Travel Booking Website

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Prepared By	Project Team
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Table of Contents

1. Introduction
2. Overall Description
3. System Features
4. Functional Requirements
5. External Interface Requirements
6. Non-Functional Requirements
7. System Models
8. Constraints, Assumptions and Dependencies
9. Testing and Acceptance Criteria
10. Future Enhancements
11. Appendix

Revision History

Version	Date	Description	Author
0.1	01 Oct 2026	Initial outline	Project Team
1.0	08 Oct 2026	First complete draft	Project Team

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the requirements of the **MakeMyTrip Clone**, a frontend web application inspired by the MakeMyTrip travel booking platform. It defines the functional and non-functional requirements, interfaces, constraints and acceptance criteria so that developers, testers, reviewers and instructors share a common understanding of the system.

1.2 Scope

The MakeMyTrip Clone is a simple, attractive and user-friendly travel booking interface where users can search and explore travel services such as flights, hotels and other travel options. The project focuses on **frontend design and user experience**. It uses static or mock data and does **not** include real-time booking, payment processing, or a production back-end.

1.3 Objectives

- Understand and apply core frontend concepts using HTML, CSS and JavaScript.
- Recreate a real-world travel booking website interface with a modern look and feel.
- Provide interactive search forms, tabs, sliders and validation using JavaScript.
- Deliver a responsive layout that works on desktop, tablet and mobile devices.

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
UI / UX	User Interface / User Experience
HTML	HyperText Markup Language - structure of web pages
CSS	Cascading Style Sheets - styling and layout
JS	JavaScript - interactivity and dynamic behaviour
DOM	Document Object Model
OTW	One-Way trip
RT	Round Trip

1.5 Intended Audience

- **Developers** - to implement the features described.
- **Testers** - to derive test cases.
- **Instructors / Reviewers** - to evaluate scope and completeness.
- **Users** - to understand what the system offers.

1.6 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- MDN Web Docs - HTML, CSS and JavaScript references (developer.mozilla.org).
- MakeMyTrip public website (used for visual and functional inspiration only).

2. Overall Description

2.1 Product Perspective

The product is a standalone, client-side website that runs entirely in a web browser. It is not connected to any real airline, hotel or payment system. All listings, prices and offers are sourced from local mock data (JavaScript arrays / JSON).

2.2 Product Functions (Summary)

- Top navigation bar with logo, service links and Login / Sign-up option.
- Travel search section with tabs for Flights, Hotels, Holiday Packages, Trains, Buses and Cabs.
- Flight search form: trip type, from, to, departure/return date, travellers and class.
- Hotel search form: city, check-in/check-out dates, rooms and guests.
- Offers and deals carousel; popular destinations section.
- Search results page showing mock flights or hotels with filters and sorting.
- Footer with quick links, support information and social icons.

2.3 User Classes and Characteristics

User Class	Description	Technical Skill
Guest Visitor	Browses the site, searches flights/hotels, views offers and destinations.	Basic
Registered User (simulated)	Uses the login / sign-up forms (front-end validation only).	Basic
Administrator / Developer	Edits mock data and source files to update content.	Intermediate

2.4 Operating Environment

- Modern browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari (latest two versions).
- Devices: desktops, laptops, tablets and smartphones.
- No server or database required; can be opened locally or hosted on any static host (e.g. GitHub Pages).

2.5 Design and Implementation Constraints

- Only HTML, CSS and JavaScript may be used (frameworks are optional and not required).
- No real payment gateway or booking API is integrated.
- Images and icons must be free to use or self-created.

3. System Features

Feature	Description	Priority
3.1 Navigation Bar	Provides access to all main sections of the website.	High
3.2 Travel Search Section	Tabbed search panel for flights, hotels and other services.	High
3.3 Search Results	Displays mock results with filters and sorting.	High
3.4 Offers and Deals	Carousel of promotional offers with coupon codes.	Medium
3.5 Popular Destinations	Cards showing trending destinations with images and prices.	Medium
3.6 Login / Sign-up	Modal forms with client-side validation.	Medium
3.7 Responsive Design	Layout adapts to different screen sizes.	High
3.8 Footer and Support	Quick links, contact and social media information.	Low

4. Functional Requirements

Each requirement is uniquely identified (FR-xx) for traceability. The keyword **shall** denotes a mandatory requirement and **should** a recommended one.

4.1 Navigation

ID	Requirement
FR-01	The system shall display a fixed/sticky navigation bar with the site logo and links to Flights, Hotels, Holidays, Trains, Buses and Cabs.
FR-02	The system shall provide Login / Sign-up buttons that open a modal window.
FR-03	On small screens the navigation bar shall collapse into a hamburger menu.

4.2 Flight Search

ID	Requirement
FR-04	The system shall allow the user to choose trip type: One Way, Round Trip or Multi-City.
FR-05	The system shall provide From and To fields with city auto-suggestions from a local list.
FR-06	The system shall provide a swap button to interchange the From and To cities.
FR-07	The system shall provide date pickers for departure and return; past dates shall be disabled.
FR-08	The system shall allow selecting number of travellers (adults, children, infants) and travel class.
FR-09	The system shall validate that From and To are not the same and that all required fields are filled before searching.
FR-10	On clicking Search, the system shall display matching mock flight results.

4.3 Hotel Search

ID	Requirement
FR-11	The system shall accept city/hotel name, check-in date, check-out date, rooms and guests.
FR-12	The system shall ensure check-out date is later than check-in date.
FR-13	The system shall display mock hotel cards with image, name, rating, price and amenities.

4.4 Results, Filters and Sorting

ID	Requirement
FR-14	The system shall display flight results with airline, timings, duration, stops and price.
FR-15	The system shall allow filtering by price range, airline/star rating and stops.
FR-16	The system shall allow sorting by price (low to high / high to low) and duration.
FR-17	The system should display a 'No results found' message when no data matches.
FR-18	Clicking 'Book Now' shall show a confirmation message stating this is a demo (no real booking).

4.5 Offers, Destinations and Other Sections

ID	Requirement
FR-19	The system shall display an auto-scrolling offers carousel with previous/next controls.
FR-20	The system shall allow users to copy a coupon code from an offer card.
FR-21	The system shall display popular destination cards with hover effects.
FR-22	The system should provide a newsletter / subscribe input with email validation.

4.6 Login and Sign-up

ID	Requirement
FR-23	The system shall validate email format and password length (minimum 8 characters).
FR-24	The system shall display clear error messages for invalid input.
FR-25	The system may store a simulated session using browser localStorage.

5. External Interface Requirements

5.1 User Interface

- Clean layout with a consistent colour scheme (red, white and dark blue accents).
- Hero banner with the search card overlaid; clearly visible call-to-action buttons.
- Consistent typography, spacing and iconography across all sections.
- Visual feedback: hover effects, focus outlines, loading animation on search.

5.2 Hardware Interfaces

No special hardware is required. Standard input devices (mouse, keyboard, touch screen) are supported.

5.3 Software Interfaces

- Web browser with HTML5, CSS3 and ES6 JavaScript support.
- Optional: Font Awesome / Google Fonts via CDN for icons and typography.

5.4 Communication Interfaces

The application does not require network communication except for loading optional CDN assets. No back-end API calls are made.

6. Non-Functional Requirements

Attribute	Requirement
Performance	Home page shall load within 3 seconds on a standard broadband connection; interactions shall respond within 300 ms.
Usability	A first-time user shall be able to complete a search within 1 minute without instructions.
Responsiveness	Layout shall adapt to widths from 320 px (mobile) to 1920 px (desktop).
Compatibility	Shall work on the latest two versions of Chrome, Firefox, Edge and Safari.
Accessibility	Shall use semantic HTML, alt text for images, sufficient colour contrast and keyboard navigation.
Maintainability	Code shall be modular, commented and organised into separate HTML, CSS and JS files.
Reliability	Invalid input shall never break the page; errors shall be handled gracefully.
Security	Inputs shall be sanitised before rendering; no sensitive data shall be stored in plain form.
Portability	Shall run without installation from local files or any static web host.

7. System Models

7.1 Page Structure

- **index.html** - Home page (navbar, hero + search, offers, destinations, footer).
- **results.html** - Flight / hotel search results page.
- **css/style.css** - Global styles and responsive rules.
- **js/script.js** - Search logic, validation, carousel and modal handling.
- **data/data.js** - Mock data for cities, flights, hotels and offers.
- **images/** - Banner, destination and icon assets.

7.2 User Flow

User opens Home page → selects service tab → enters search details → form validation → (valid) results page with filters → selects an option → demo confirmation message. If validation fails, an error message is shown and the user stays on the form.

7.3 Use Cases

Use Case	Actor	Description	Outcome
UC-01 Search Flights	Visitor	Enter route, dates and travellers, then search.	Matching flights displayed.
UC-02 Search Hotels	Visitor	Enter city, dates and guests, then search.	Matching hotels displayed.
UC-03 Filter / Sort	Visitor	Apply filters or sorting on results.	Updated result list.
UC-04 View Offers	Visitor	Browse and copy coupon codes.	Coupon copied.
UC-05 Login / Sign-up	Visitor	Submit credentials in modal.	Validation / simulated login.

8. Constraints, Assumptions and Dependencies

8.1 Constraints

- No real transactions, payments or live inventory are supported.
- Time-bound academic project with limited scope.

8.2 Assumptions

- Users have a modern browser with JavaScript enabled.
- Mock data is sufficient to demonstrate all features.

8.3 Dependencies

- Optional CDN resources (Font Awesome, Google Fonts).
- Image assets stored locally in the project folder.

9. Testing and Acceptance Criteria

The system will be considered complete when all high-priority functional requirements are implemented and the following tests pass.

Test ID	Scenario	Expected Result
T-01	Search flights with same From and To city	Error message is displayed; search is blocked.
T-02	Select a past departure date	Past dates are disabled / rejected.
T-03	Swap From and To	Values are interchanged correctly.
T-04	Apply price filter on results	Only flights in selected range are shown.
T-05	Resize browser to 360 px width	Layout adapts; no horizontal scrolling.
T-06	Submit login form with invalid email	Validation error is displayed.
T-07	Open site in Chrome, Firefox, Edge, Safari	UI and features work consistently.

10. Future Enhancements

- Integrate real flight and hotel APIs for live data.

- Add a Node.js / Express back-end with a database (MongoDB / MySQL).
- Implement real authentication, user profiles and booking history.
- Integrate a payment gateway (test mode).
- Add multi-language support, dark mode and Progressive Web App (PWA) features.
- Add maps, reviews and price-alert notifications.

11. Appendix

A. Suggested Folder Structure

```
makemytrip-clone/  
|-- index.html  
|-- results.html  
|-- css/  
|   |-- style.css  
|-- js/  
|   |-- script.js  
|-- data/  
|   |-- data.js  
|-- images/
```

B. Requirement Traceability (Summary)

Feature	Requirement IDs
Navigation Bar	FR-01 to FR-03
Flight Search	FR-04 to FR-10
Hotel Search	FR-11 to FR-13
Results, Filter, Sort	FR-14 to FR-18
Offers and Destinations	FR-19 to FR-22
Login / Sign-up	FR-23 to FR-25

Disclaimer: This project is an educational clone created for learning purposes. MakeMyTrip is a registered trademark of its respective owner. This project is not affiliated with or endorsed by MakeMyTrip.